

A100 – Lepton Sector I: The ξ -Ladder

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.1 The Ladder

[B] The lepton masses can be written as rungs of a ξ -ladder:

$$m_i = r_i \xi^{p_i} \nu,$$

with rational structure numbers r_i , rational exponents p_i , and a bridge constant ν . For the leptons these are

$$(r_e, p_e) = \left(\frac{4}{3}, \frac{3}{2}\right), \quad (r_\mu, p_\mu) = \left(\frac{16}{5}, 1\right), \quad (r_\tau, p_\tau) = \left(\frac{25}{9}, \frac{2}{3}\right).$$

[K] In the *ratios* ν cancels, leaving parameter-free expressions:

$$\frac{m_\mu}{m_e} = \frac{12}{5} \xi^{-1/2}, \quad \frac{m_\tau}{m_\mu} = \frac{125}{144} \xi^{-1/3}, \quad \frac{m_\tau}{m_e} = \frac{25}{12} \xi^{-5/6}.$$

According to A080 these are the intrinsic quantities: they depend on the geometry and nothing else. They are precisely what is to be measured against.

.2 Where the Exponents Come From

[K] The exponents p_i are not free. They are winding numbers on a *logarithmic* torus with step size $\Delta p = 1/3$. This step size is the fundamental third of the harmonic series – the same geometry of periodic boundary conditions that generates overtones on a vibrating string. Half-integer values are geometric means between two winding levels, negative values counter-windings.

[K] The three lepton exponents lie on this ladder:

$$p_e = \frac{3}{2}, \quad p_\mu = 1, \quad p_\tau = \frac{2}{3},$$

with steps

$$p_e - p_\mu = \frac{1}{2} \text{ (geometric mean),} \quad p_\mu - p_\tau = \frac{1}{3} \text{ (fundamental third).}$$

These are the same winding numbers (n_θ, n_ϕ) that also fix the prefactors $r_i = n_\phi^2/n_\theta$ (A110): mass prefactor and exponent come from one source, not two. The ladder is therefore not a fit but the reading of the resonance structure of the torus.

.3 What the Ladder Delivers

[K] The calculation against PDG values gives:

Ratio	Ladder	PDG	Deviation
m_μ/m_e	207.85	206.77	+0.52 %
m_τ/m_μ	16.992	16.817	+1.04 %
m_τ/m_e	3532	3477	+1.57 %

[K] The tolerance standard from A010 and A080 requires three figures side by side. The measurement precision of the comparison quantities is at m_μ/m_e in the range 10^{-9} , at m_τ about $7 \cdot 10^{-5}$. The determination tolerance of the ladder values is orders of magnitude coarser: the structure numbers are declared rational, not determined to significant figures.

Assessment: the ladder carries the *hierarchy* and the per cent level. It does not carry precision. A one per cent deviation is not a finding against the structure here, because the structure claims nothing more precise at this level – but it is also not a success that could be reported to five figures.

.4 The Reverse Direction: What the Data Return for ξ

[K] Solving the three ratios for ξ , the data give

$$\xi_{\mu/e} = 1.347 \cdot 10^{-4}, \quad \xi_{\tau/\mu} = 1.375 \cdot 10^{-4}, \quad \xi_{\tau/e} = 1.358 \cdot 10^{-4},$$

against $4/30000 = 1.3333 \cdot 10^{-4}$, i.e. +1.1 to +3.2 %. The three values scatter among themselves by about 2 %.

This is the honest empirical status: the ladder is consistent in the ratios at the per cent level, not exactly. It does not return a uniform ξ .

[K] The same from the other side: solving for the bridge constant per lepton gives 248.9, 247.6, and 245.1 GeV, scattered around 246.22 GeV. The residual deviation is therefore absorbed into the bridge constant and does *not* cancel completely in the ratios.

.5 The Countercheck That Fails

[K] The obvious conjecture, that the residual deviations follow the fractal correction factor, is *denied* by the calculation. $K_{\text{frak}} = 1 - 100\xi$ corresponds to -1.333% , the half and three-half powers to -0.665 and -1.995% . The measured deviations are $+0.52$ and $+1.04\%$ – they fit neither in magnitude nor in *sign* a single K_{frak} power.

This section is here because a failed countercheck belongs to the finding just as much as a successful one. The obvious explanatory attempt has been tested and has failed; it is not rescued by a free exponent.

The striking doubling is arithmetically forced – and does not evidence a correction.

The τ/μ deviation is almost exactly double the μ/e deviation (1.04 % vs. 0.52 %), and the τ/e deviation is their sum ($0.52 + 1.04 = 1.56$ %). This is not a new law and not a coincidence but a pure bookkeeping necessity: along a multiplicative ladder in which $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ holds, the logarithmic deviations add. Of the three deviations therefore only *one* is independently measured (the ratio 1.98); the third is fixed as a sum.

Importantly, this step does *not* establish anything about the fractal factor that the countercheck just rejected – on the contrary, it shows that the doubling singles out no correction at all: it already follows from the ladder form alone. One who reads a law from the doubling reads a single measured number three times.

[K] The factor that carries the ladder per generation (its generation-counting step, order $N_0 \approx 39$) is a *different* quantity from the fixed scale 100 in K_{frak} : the one is the ladder step, the other the RG scale of the absolute correction (A040). The value of this ladder step is not derived – that is the one open point named below.

.6 Two Mutually Exclusive Computation Modes

[STIPULATION] Before any of these deviations is booked, the computation mode must be fixed. There are two, and they exclude each other.

Mode 1, reference-free. All ratios are computed purely from ξ , without anchor. Then all three stand independently against the data, each 0.5 to 1.6 % off, and all three are legitimate *check points*.

A note on terminology, because otherwise a contradiction with A010 arises: these three ratios are *not* prediction quantities of the series. The prediction list comprises four quantities, and the lepton masses appear there solely via the τ -mass from the circulant (A110), not via the ladder. The ladder provides check points at the per cent level – less than a prediction and more than nothing.

Mode 2, anchored. m_μ/m_e is declared a reference point. Then this ratio is *spent*: its deviation may no longer be booked as a test result, otherwise the same quantity is used twice – once as anchor, once as test.

[B] Mixing the two modes is the most common error in this sector. The A-series consistently uses mode 1, unless something else is explicitly stated.

.7 Verification Script

- All numbers in this document – ladder ratios, reverse- ξ , bridge constants per lepton, the failed K_{frak} countercheck: `python/A_Serie_Skripte/a100_leiter.py`

.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$r_i \in \mathbb{Q}$	Rational prefactor in $m_i = r_i \cdot \xi^{p_i} \cdot v$
$p_i \in \mathbb{Q}$	Exponent; ladder step size $\Delta p = 1/3$
(n, l, j)	Principal, orbital, total angular momentum of the mode
$v = 246 \text{ GeV}$	Higgs vacuum expectation value as energy unit

.9 Open Questions

First, the per cent scatter is intended, not a defect. The ladder is a statement at the per cent level; a ξ back-computed from it scatters accordingly by 2 %. For this reason the sector anchors the absolute scale not in the ladder but in a single reference point (A110): the ladder carries the hierarchy, the reference point the precision.

Second, the exponents p_i are derived, not merely motivated. They are winding numbers on the logarithmic torus (above), with step size $1/3$ and geometric means at half-integer values – the same source from which the prefactors r_i follow. The condition from A020 is thereby met: the exponent is fixed independently, the ξ power is a statement.

Third, the doubling is clarified. It is arithmetically forced (above): along a multiplicative ladder the logarithmic deviations add, only one is independent. No unbooked residual remains.

Fourth, exactly one point remains open – and it is named. The one independently measured deviation corresponds to a generation-counting ladder step of order $N_0 \approx 39$, whose value is not derived. It is to be separated from the fixed RG scale 100 in K_{frak} (A040) and is not explained by the bridge constant in which the residual deviation is absorbed (A080). The value of this step and the candidates for it are collected in A105; whether structure sits behind it is the only genuinely open question of the sector.

.10 Supersedes

This document subsumes: Dok. 006 (mass formula), Dok. 046 (mass calculation), Dok. 016 (particle masses), Dok. 189 (winding numbers of exponents), Dok. 190 (R55 additivity, P42 reference point), and the ladder sections from Dok. 292. Incorporated: K2 (exponent p_r), P40 (level separation), P42 (reference point), P35, R55 (evidential breadth: only one point independently measured).

.11 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.